

Advanced Machine Learning and Artificial Intelligence

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@ Data Science UAB Summer Camp

From “Decision Trees to ChatGPT”

Advanced Machine Learning and Artificial Intelligence



AI



LEARNING



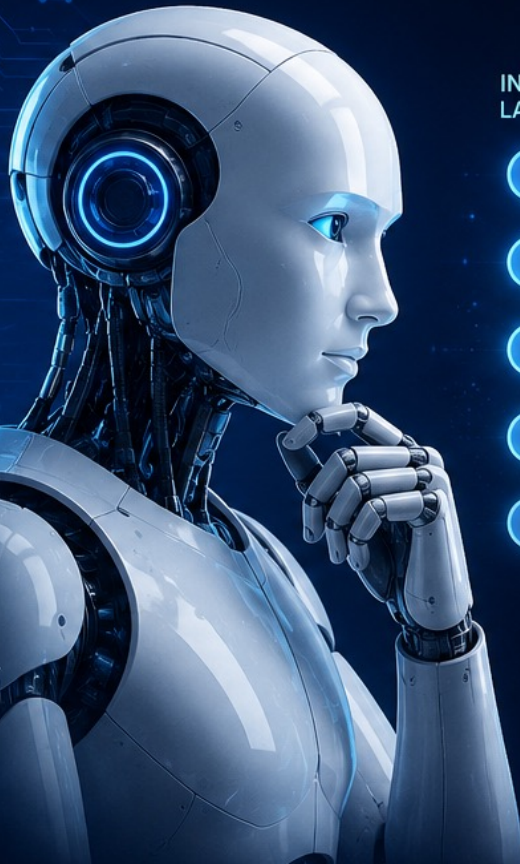
DATA



INSIGHT



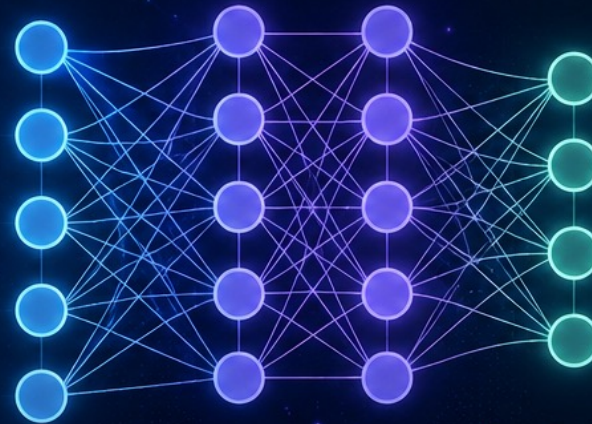
INNOVATION



INPUT
LAYER

HIDDEN LAYERS

OUTPUT
LAYER



DATA



LEARN



PREDICT



ChatGPT



What is machine learning?



Machine learning is a type of artificial intelligence that enables computers to learn from data and make predictions or decisions without being explicitly programmed.



Can you give an example?

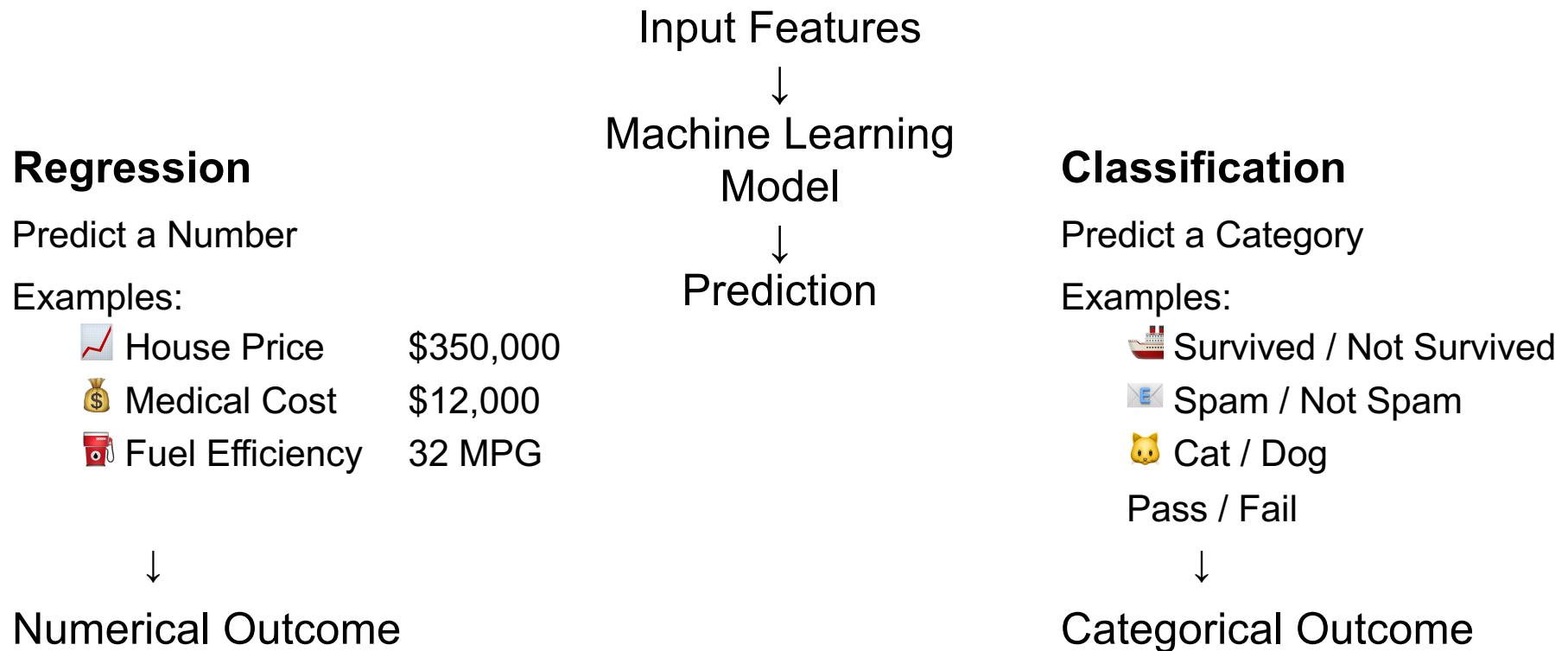


Sure! Email spam filtering is an example. The system learns from many emails to distinguish spam from important messages.

Message ChatGPT...



What Is Classification?



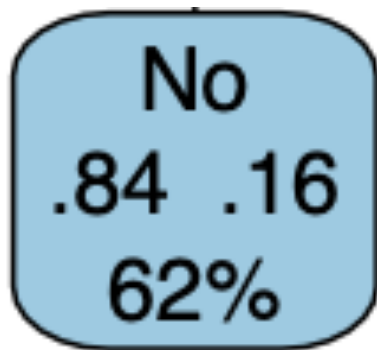
The Titanic Survival Problem

PassengerId	100
Survived	Can We Predict Who Survived?
Pclass	2
Name	Kantor, Mr. Sinai
Sex	male
Age	34
SibSp	1
Parch	0
Ticket	244367
Fare	26
Cabin	
Embarked	S

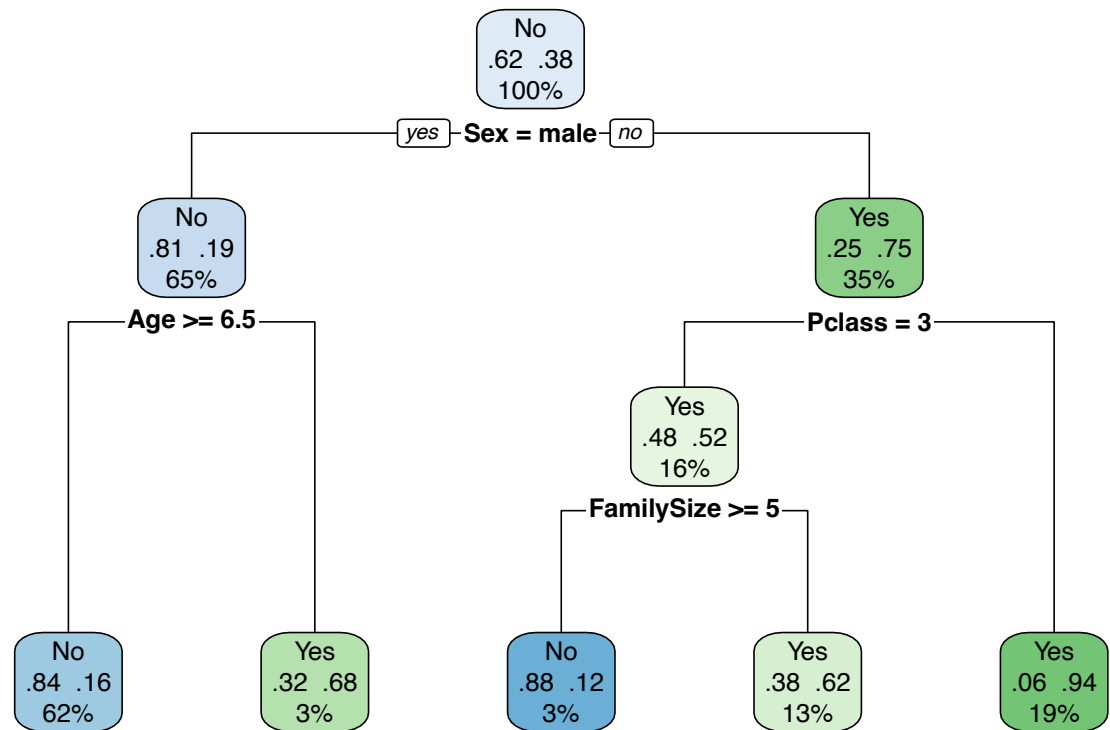
Variable	Description
survival	Survival
pclass	Passenger Class
name	First and Last Name
sex	Sex
age	Age
sibsp	Number of Siblings/Spouses Aboard
parch	Number of Parents/Children Aboard
ticket	Ticket Number
fare	Passenger Fare
cabin	Cabin
embarked	Port of Embarkation

Decision Trees

PassengerId	100
Survived	0
Pclass	2
Sex	male
Age	34
SibSp	1
Parch	0



Decision Tree for Titanic Survival

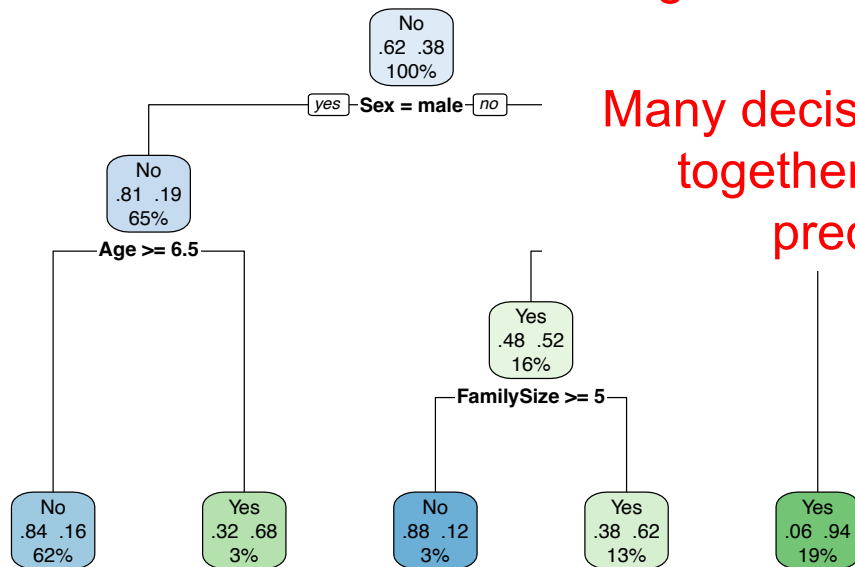


Decision Trees make predictions by asking a sequence of questions.

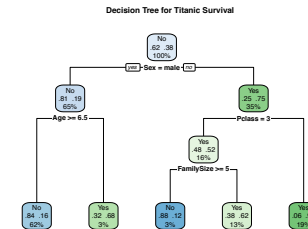
Random Forest

One Decision Tree

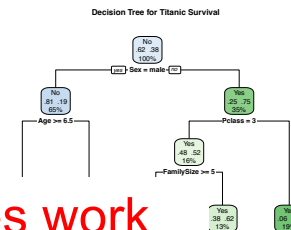
Decision Tree for Titanic Survival
Could this tree be wrong?



Many decision trees work together to make a prediction.

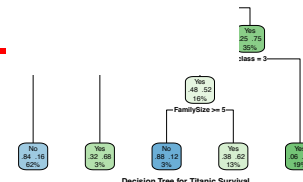


→ Survived

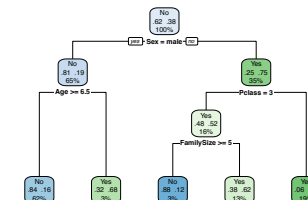


→ Survived

→ Survived



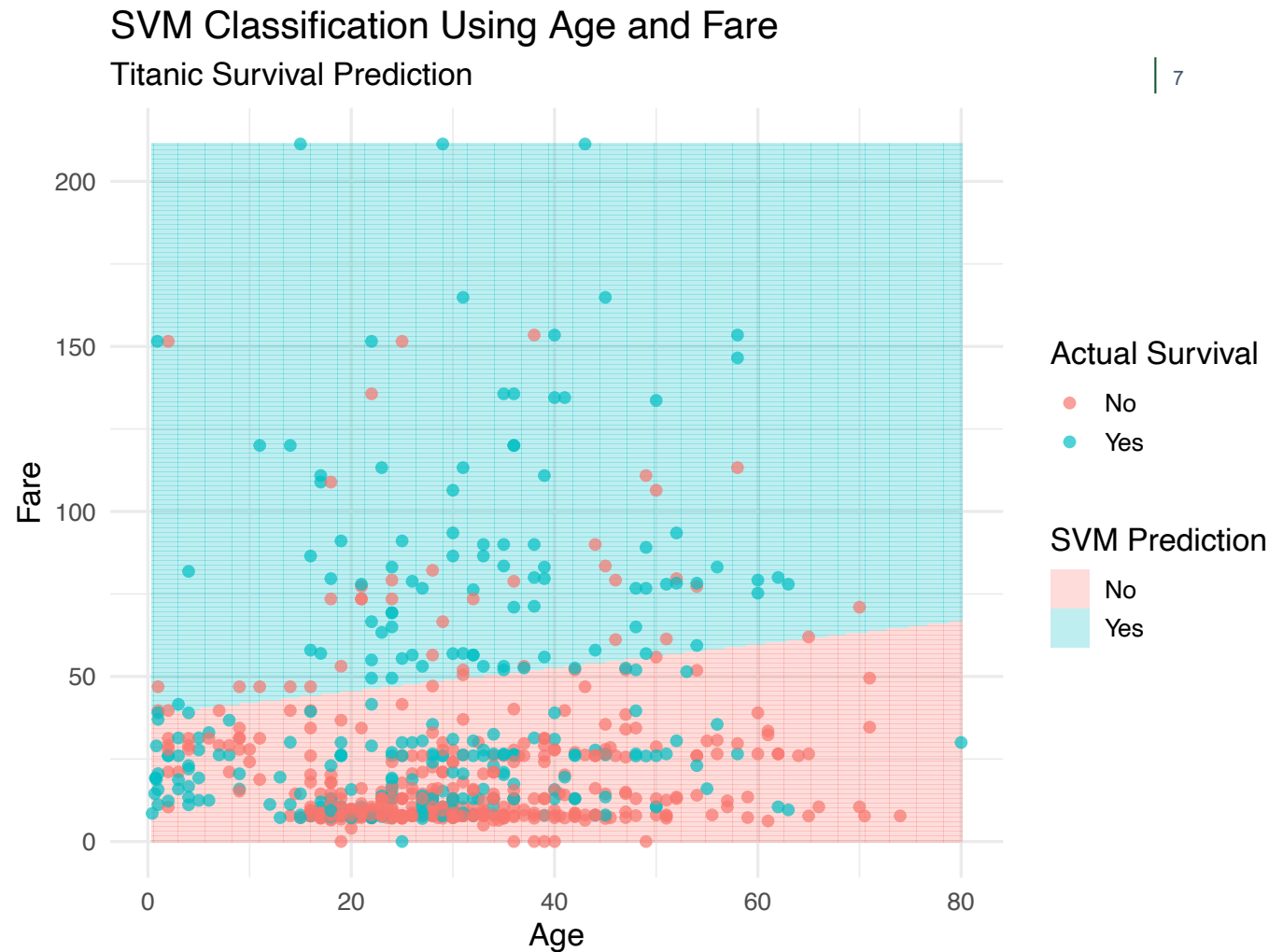
→ Not Survived



→ Survived

Support Vector Machines (SVM)

SVM finds the boundary that best separates different categories.



Comparing Classification Methods

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Decision Tree

- Idea:
Ask Questions
- Strength:
Easy to Understand
- Weakness:
Can Overfit

Random Forest

- Idea:
Many Trees Vote
- Strength:
Usually More Accurate
- Weakness:
Harder to Interpret

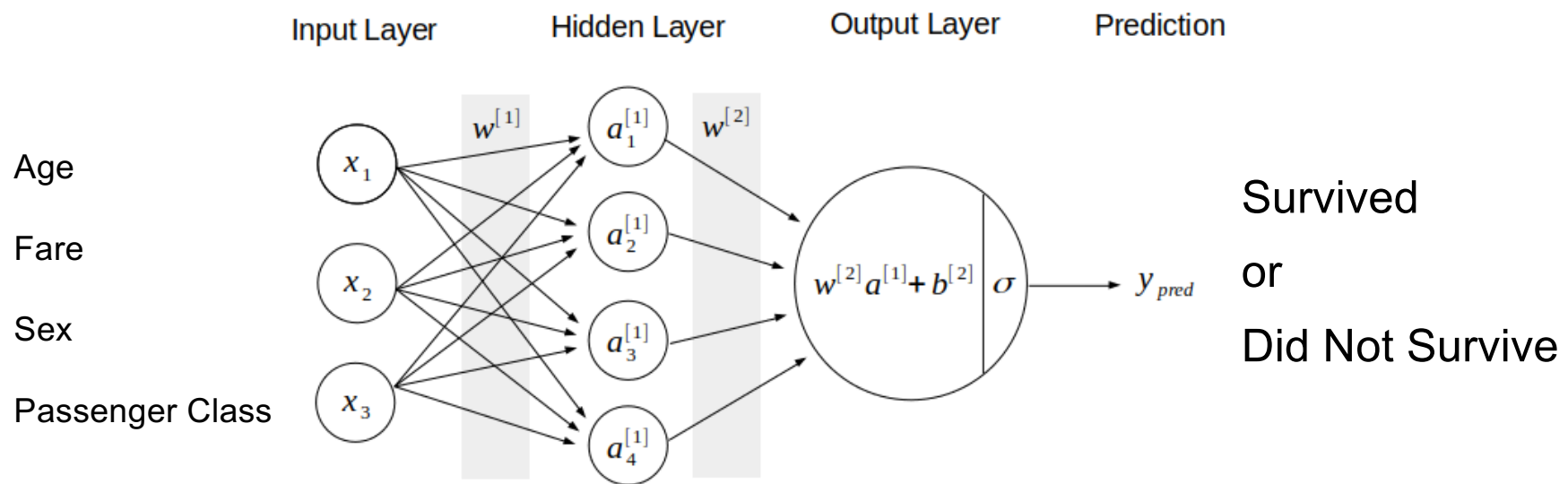
SVM

- Idea:
Find Best Boundary
- Strength:
Powerful Classifier
- Weakness:
Less Transparent

Which Method Is Best?

Neural Networks

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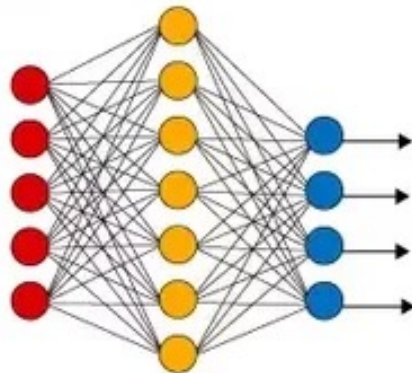
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Neural Networks learn complex patterns from large amounts of data.

Deep Learning

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Simple Neural Network



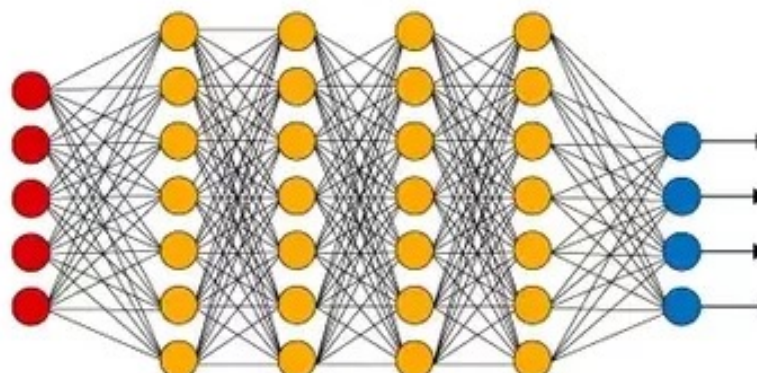
● Input Layer

● Hidden Layer

● Output Layer

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Deep Learning Neural Network



Applications:



Image Recognition



Speech Recognition



Translation



ChatGPT

Deep Learning = Neural Networks with Many Layers

Large Language Models and ChatGPT

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LLM Pipeline

Massive Text Data



Deep Neural Network



Learn Language Patterns



Predict Next Word

ChatGPT does not search for answers.

It predicts likely words based on patterns learned from enormous amounts of text.

- ChatGPT:

A random forest is a collection of decision trees that vote together to make predictions.

```
library(ellmer)
```

```
chat <- chat_ollama(model = "llama3.2")
```

```
chat$chat("Explain a random forest.")
```

What AI Can and Cannot Do

| 12

Can

Cannot